# BEFA UNIT III

## PRODUCTION

Production is the transformation or conversion of resources into commodities over time. Economists view production as an activity through which utility is created or enhanced for a product. A firm is a business unit which undertakes the activity of transforming inputs into output of goods and services.


## FACTORS OF PRODUCTION

Factors of production is an economic term that describes the inputs that are used in the production of goods or services in order to make an economic profit. The factors of production include land, labor, capital and entrepreneurship. These production factors are also known as management, machines, materials and labor, and knowledge has recently been talked about as a potential new factor of production.

### 1. Land

Land is short for all the natural resources available to create supply. It includes raw land and anything that comes from the land. It can be a non-renewable resource.

That includes commodities such as oil and gold. It can also be a renewable resource, such as timber. Once man changes it from its original condition, it becomes a capital good. For example, oil is a natural resource, but gasoline is a capital good. Farmland is a natural resource, but a shopping center is a capital good.

The income earned by owners of land and other resources is called rent.

### 2. Labour

Labor is the work done by people. The value of labor depends on workers' education, skills, and motivation. It also depends on productivity. That measures how much each hour of worker time produces in output.

The reward or income for labor is wages.

### 3. Capital

Capital is short for capital goods. These are man-made objects like machinery, equipment, and chemicals, that are used in production. That's what differentiates them from consumer goods. For example, capital goods include industrial and commercial buildings, but not private housing. A commercial aircraft is a capital good but a private jet is not.

The income earned by owners of capital goods is called interest.

### 4. Entrepreneurship

Entrepreneurship is the drive to develop an idea into a business. An entrepreneur combines the other three factors of production to add to supply. The most successful are innovative risk-takers.

The income entrepreneurs earn is profits.

## PRODUCTION FUNCTION

The production function expresses a functional relationship between physical inputs and physical outputs of a firm at any particular time period. The output is thus a function of inputs. So, production function is an input – output relationship. Mathematically production function can be written as

$$Q= f\,(L_1,L_2\ C,O,T)$$

Q = Output
f = Function of
$L_1$ = Land
$L_2$ = Labour
C = Capital
O = Organization
T = Technology

Here output is the function of inputs. Hence output becomes the dependent variable and inputs are the independent variables.

**Definition :**

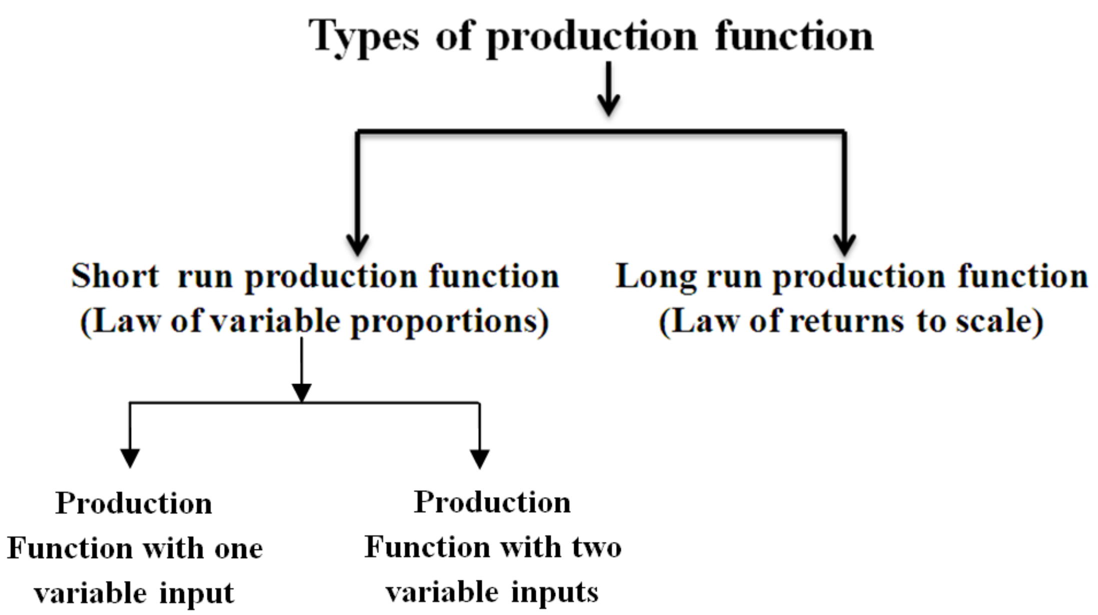
Samueson defines the production function as "The technical relationship which reveals the maximum amount of output capable of being produced by each and every set of inputs"

Michael R Baye defines the production function as" That function which defines the maximum amount of output that can be produced with a given set of inputs."

**Assumptions:**

Production function has the following assumptions.

1. The production function is related to a particular period of time.
2. There is no change in technology.
3. The producer is using the best techniques available.
4. The factors of production are divisible.
5. Production function can be fitted to a short run or to long run.



# BEFA UNIT III

## PRODUCTION FUNCTION WITH ONE VARIABLE INPUT

The law of variable proportions which was earlier called as "Law of diminishing returns has played a vital role in the modern economics theory. Assume that a firms' production function consists of fixed quantities of all inputs (land, equipment, etc.) except labour which is a variable input. If you go on adding the variable input, say, labor, the total output in the initial stages will increase at an increasing rate, and after reaching certain level of output the total output will increase at declining rate. If variable factor inputs are added further to the fixed factor input, the total output may decline. This law is of universal nature and it proved to be true in agriculture.

**Assumptions of the Law**: The law is based upon the following assumptions:

1. Only one factor is varied
2. The scale of output is unchanged
3. The technique of production is unchanged
4. All units of factor input varied are homogeneous

**Three stages of law:**

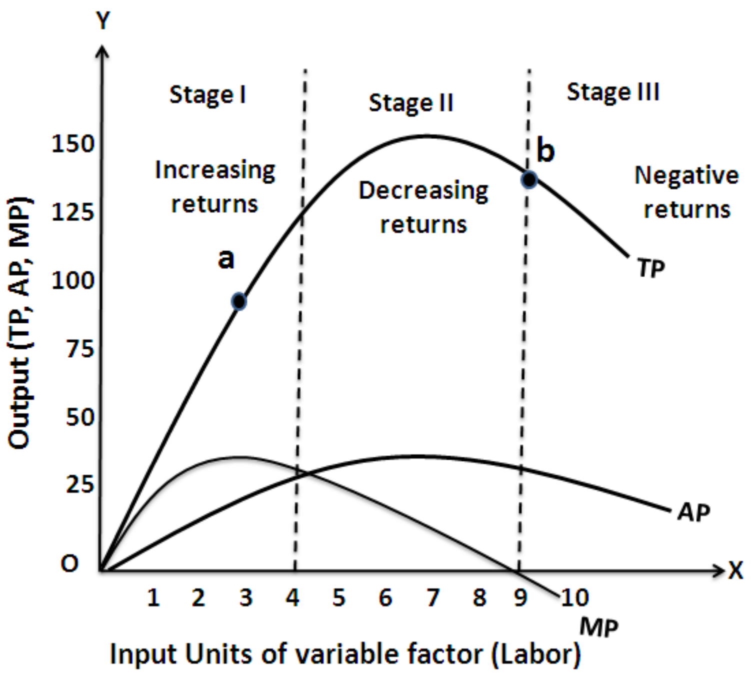
The behaviour of the Output when the varying quantity of one factor is combined with a fixed quantity of the other can be divided in to three district stages. The three stages can be better understood by following the table.

**Production function with one variable input**

| Fixed Factor (FF) (land) | Variable Factor (VF) (labor) | Total Product (TP) | Average Product (AP) | Marginal Product (MP) | |
|---|---|---|---|---|---|
| 1 | 1 | 20 | 20 | 20 | STAGE I |
| 1 | 2 | 50 | 25 | 30 | |
| 1 | 3 | 90 | 30 | 40 | |
| 1 | 4 | 120 | 30 | 30 | STAGE II |
| 1 | 5 | 135 | 27 | 15 | |
| 1 | 6 | 144 | 24 | 9 | |
| 1 | 7 | 147 | 21 | 3 | |
| 1 | 8 | 148 | 18.5 | 1 | |
| 1 | 9 | 148 | 16.4 | 0 | |
| 1 | 10 | 145 | 14.5 | - 3 | STAGE III |

To clarify the relationship, the following are measurements of product.

1. **Total product(TP)**:- Means the total number of units of output produced per unit of time by all factor inputs.
2. **Average product(AP)**:- Is obtained by dividing the total product by the total units of variable factor.
3. **Marginal product(MP):-** is defined as the change in the total product per unit change in the variable input.




| | I | II | III |
|---|---|---|---|
| **TP** | Increases at an increasing rate | Increases at a decreasing rate and becomes maximum | Decreases |
| **AP** | Increases and reaches maximum | Decreases | Continuous to decrease |
| **MP** | Increases and reaches maximum | Continuous to fall | Becomes negative |

From the above graph the law of variable proportions operates in three stages. In the first stage, total product increases at an increasing rate. The marginal product in this stage increases at an increasing rate resulting in a greater increase in total product. The average product also increases. This stage continues up to the point where average product is equal to marginal product. The law of increasing returns is in operation at this stage. The law of diminishing returns starts operating from the second stage onwards. At the second stage total product increases only at a diminishing rate. The average product also declines. The second stage comes to an end where total product becomes maximum and marginal product becomes zero. The marginal product becomes negative in the third stage. So the total product also declines. The average product continues to decline.

We can sum up the above relationship thus when 'AP' is rising, "MP' rises more than "AP; When 'AP" is maximum and constant, 'MP' becomes equal to 'AP' when 'AP' starts falling, 'MP' falls faster than 'AP'.

Thus, the total product, marginal product and average product pass through three phases, viz., increasing diminishing and negative returns stage. The law of variable proportion is nothing but the combination of the law of increasing and demising returns.

## PRODUCTION FUNCTION WITH TWO VARIABLE INPUTS

Isoquants analyse and compare the different combinations of capital & labour and output. The term isoquant has its origin from two words "iso" and "quantus". 'iso' is a Greek word meaning 'equal' and 'quantus' is a Latin word meaning 'quantity'. Isoquant therefore, means equal quantity. An isoquant curve is therefore called as iso-product curve or equal product curve or production indifference curve.

Thus, an isoquant shows all possible combinations of two inputs, which are capable of producing equal or a given level of output. Since each combination yields same output, the producer becomes indifferent towards these combinations.

**Assumptions:**

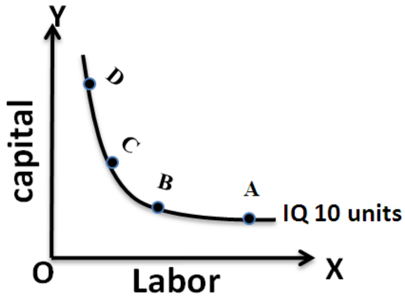
1. There are only two factors of production, viz. labour and capital.
2. The two factors can substitute each other up to certain limit
3. The shape of the isoquant depends upon the extent of substitutability of the two inputs.
4. The technology is given over a period.

**For example**:- Now the firm can combine labor and capital in different proportions and can maintain specified level of output say, 10 units of output of a product X. It may combine alternatively as follows:

In the below table, combination 'A' represent 1 unit of capital and 10 units of labour and produces '10' units of a product. All other combinations in the table are assumed to yield the same given output of a product say '10' units by employing any one of the alternative combinations of the two factors labour and capital. If we plot all these combinations on a paper and join them, we will get a curve called Iso-quant curve as shown below.

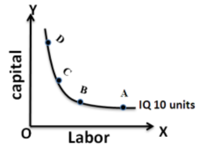
Labour is on the X-axis and capital is on the Y-axis. IQ is the Iso-Quant curve which shows all the alternative combinations A, B, C, D which can produce 10 units of a product.

| Combination | Capital | Labor | output |
|---|---|---|---|
| A | 1 | 10 | 10 units |
| B | 2 | 6 | 10 " |
| C | 3 | 3 | 10 " |
| D | 4 | 1 | 10 " |

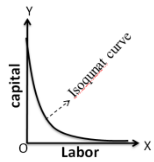



**Features of an ISOQUANT:**

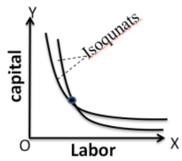
1. **Downward sloping:-**If one of the inputs is reduced, the other input has to be increased. There is no question of increase in both the inputs to yield a given output.



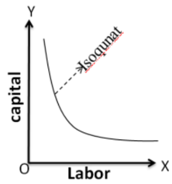

2. **Don't touch the axes:-** The isoquant touches neither X-axis nor Y-axis, as both inputs are required to produce a given product.If an isoquant is touching the X-axis, it means output is possible even by using a factor(Ex: Labor alone without using capital). But, this is unrealistic.



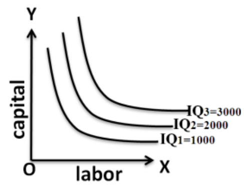

3. **Don't intersect:-** Iso-quants representing different levels of output never intersect or touch or be tangent to each other. If they intersect to each other, they have a common point on them which means that the same amount of labor and capital produce two different levels of output.




4. **Convex to origin:-**Isoquants are convex to the origin. It is because the inputs factor are not perfect substitutes. One input factor is substituted by other input factor in a decreasing marginal rate.The convexity of isoquant suggests that MRTS is diminishing which means that as quantities of one factor-labor is increased , the less of another factor-capital will be given up, if output level is to be kept constant.




5. **Upper isoquants represent higher level of output**:- Each isoquant represents a different quantity of output. Higher isoquants indicate a higher level of output.




## LAW OF RETURNS TO SCALE

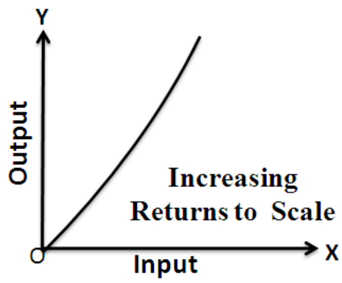
The concept of variable proportions is a short-run phenomenon as in these period fixed factors cannot be changed and all factors cannot be changed. On the other hand in the long-term all factors can be changed as made variable. When we study the changes in output when all factors or inputs are changed, we study returns to scale. An increase in the scale means that all inputs or factors are increased in the same proportion. In variable proportions, the cooperating factors may be increased or decreased and one faster (Ex. Land in agriculture (or) machinery in industry) remains constant so that the changes in proportion among the factors result in certain changes in output. In returns to scale, all the necessary factors or production are increased or decreased to the same extent so that whatever the scale of production, the proportion among the factors remains the same.

**Assumptions**

1. Technique of production is unchanged
2. All units of factors are homogeneous
3. Returns are measured in physical terms

When a firm expands, its scale increases all its inputs proportionally, then technically there are three possibilities.

**1. Law of increasing returns to scale:-** if a proportionate/percentage increase in the output is larger than the proportionate/percentage increase in inputs, there are increasing returns.




**For example:** If a 5% increase in inputs, results in 10% increase in the output, a firm is said to attain increased returns.

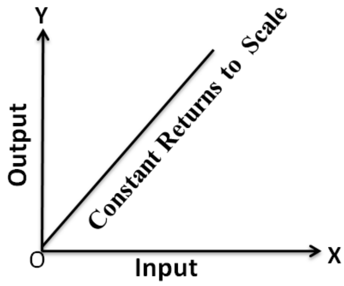
$$\text{Production Factor Co-efficient} = \frac{\text{\% Change in output}}{\text{\% Change in input}}$$

If PFC > 1, it means increasing returns to scale

If PFC = 1, it means constant returns to scale

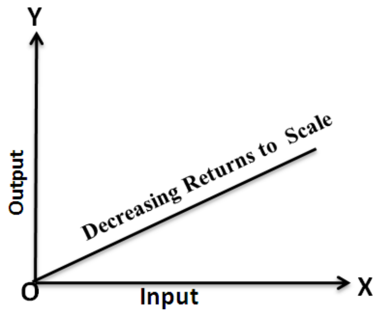
If PFC < 1, it means decreasing returns to scale

**2. Law of constant returns to scale:-** if the proportionate increase in all the inputs is equal to the proportionate increase in output, then situation of constant returns to scale occurs.




**For Example**:- If the inputs are increased at 10% and if the resultant output also increases a 10% then the organization is said to achieve constant returns to scale.

**3. Law of decreasing returns to scale:-** if the proportionate increase in output is less than the proportionate increase in input, then a situation of decreasing returns to scale occurs.




**For Example**:- If the inputs are increased by 10% and if the resultant output increases only by 5% then the organization is said to achieve decreasing returns to scale.

the following table illustrates these laws clearly.

| Capital (in units) | Labor (in units) | % of increase in both inputs | Output (in units) | % of increase in output | Laws applicable |
|---|---|---|---|---|---|
| 1 | 3 | - | 50 | - | - |
| 2 | 6 | 100 | 120 | 140 | Law of increasing returns to scale |
| 4 | 12 | 100 | 240 | 100 | Law of constant returns to scale |
| 8 | 24 | 100 | 360 | 50 | Law of decreasing returns to scale |

From the above table it is clear that with 1 unit of capital and 3 units of labor, the firm produces 50 units of output. When the inputs are doubled 2 units of capital and 6 units of labor, the output has gone up to 120 units. Thus when inputs are increased by 100%, the output has increased by 140%. That is , output has increased by more than double. This is governed by **law of increasing returns to scale**.

When the inputs are further doubled that is to 4 units of capital and 12 units of labor, the output has gone to 240 units. Thus, when inputs are increased by 100%., the output has increased by 100%. That is, output has doubled. This governed by **law of constant returns to scale.**

When the inputs further doubled, that is, to 8 units of capital and 24 units of labor, the output has gone up to 360 units. Thus, when inputs are increased by 100%, the output has increased by only 50%. This is governed by **law of decreasing returns to scale.**

## DIFFERENT TYPES OF PRODUCTION FUNCTIONS

### 1. Cobb-Douglas Production Function

This production function was proposed by C. W. Cobb and P. H. Dougles. This famous statistical production function is known as Cobb-Douglas production function. Originally the function is applied on the empirical study of the American manufacturing industry from 1899 to 1922. Cobb – Douglas production function takes the following mathematical form.

$$Q = aL^bK^C$$

$$Q = 1.01L^{0.75}K^{0.25}$$

**Q = Total production/Output**
**a = Total factor productivity/Multi -factor productivity**
**(The ratio of output to the sum of associated labor and capital inputs). Productivity is measure of efficiency of labor/machines etc, in converting input into output.**

$$\textbf{Factor productivity} = \frac{\textbf{Ouput}}{\textbf{Factor input}}$$

$$\textbf{Total Factor Productivity} = \frac{\textbf{Total Ouput}}{\textbf{Total inputs}}$$

**L = Index of employment of labor = Labour units**
**K = Index of employment of capital = Capital units**
**b = Output elasticity of labor (1% change in labor results in b% change in output)**
**c = Output elasticity of capital (1% change in capital results in c% change in output)**

The above production function shows that 1% change in labor input, capital remaining the same, is associated with a 0.75 percent change in output. Similarly, 1% change in capital, labor remaining the same, is associated with a 0.25 percent change in output.

**Assumptions:**

It has the following assumptions

1. The function assumes that output is the function of two factors viz. capital and labour.
2. It is a linear homogenous production function of the first degree
3. There are constant returns to scale (b+c=1)
4. All inputs are homogenous
5. There is perfect competition
6. There is no change in technology
7. Both L&K should be positive for to Q to exist. If either of these is zero, Q will be zero. This implies that both labor and capital will be combined to get output.

**Simple Cobb-Douglas example:**

$$Q = aL^bK^c$$
$$Q = 2L^{0.5}K^{0.5}$$

**Let K=100, L=25**

$$Q = 2x100^{0.5}x25^{0.5}$$
$$= 2x10x5$$
$$= 100 \text{ units}$$

**1% increase in labor results in 0.5% increase in output:**
**Labor units after 1% increase = 100+(100 x1/100) = 101 units**

$$Q = 2L^{0.5}K^{0.5}$$
$$Q = 2 \text{ x } 101^{0.5} \text{ x } 25^{0.5}$$
$$Q = 2 \text{ x } 10.0498 \text{ x } 5$$
$$Q = 100.498 = 100.50$$

$$\% \text{ increase in output} = \frac{100.50-100}{100} x100 = 0.50\%$$

2. **Leontief Production Function:**

Leontief production function, also known as **Fixed Proportion Production Function,** uses fixed proportion of inputs having no substitutability between them. It is regarded as the limiting case for constant elasticity of substitution.

**The production function can be expressed as follows:**

$$q = \text{Min}\left(\frac{z_1}{a}, \frac{z_2}{b}\right)$$

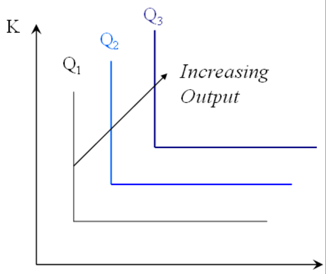
Where, q = quantity of output produced

$Z_1$ = utilized quantity of input 1

$Z_2$ = utilized quantity of input 2

a and b = constants

- Q = min{aK, bL}
- Capital and labor are perfect complements and cannot be substituted (no MRTS <=> no slope)
- Capital and labor are used in fixed-proportions
- Both inputs needed to produce output



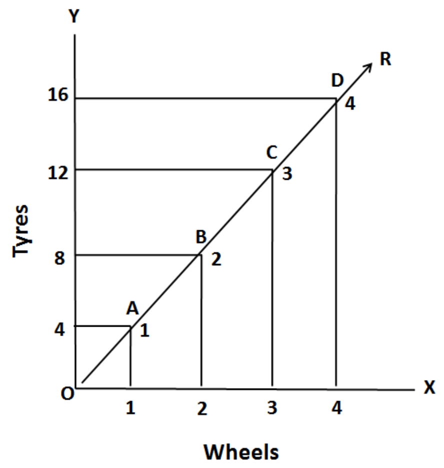

For example, tyres and steering wheels are used for producing cars. In such a case, the production function can be as follows:

Q = min ($z_1$/a, $Z_2$/b)

Q = min (number of tyres used, number of steering wheels used).

**Suppose** that the inputs "tires" and "steering wheels" are used in the production of a car (for simplicity of the example, to the exclusion of anything else). Then in the above formula $q$ refers to the number of cars produced i.e., one in our example, $z_1$ refers to the number of tires used, and $z_2$ refers to the number of steering wheels used. Assuming that each car is produced with 4 tires and 1 steering wheel, the Leontief production function is

Number of cars = Min{¼ times the number of tires, 1 times the number of steering wheels}.




In the above given figure, OR shows the fixed Tyres-Wheels ratio, if a firm wants to produce 1 car, then 4 tyres and 1 wheel must be used.

Similarly, for the production of 3 cars and 4 cars, 12 tyres and 3 wheel and 16 tyres and 4 wheel must be employed respectively.

### 3. CES Production Function

**Definition:** The **Constant Elasticity of Substitution Production Function or CES** implies, that any change in the input factors, results in the constant change in the output. In CES, the elasticity of substitution is constant and may not necessarily be equal to one or unity.

In constant elasticity of substitution production function, all the input factors are taken into the consideration such as raw material, technology, labor, capital, etc. The marginal product of one factor increases with the increase in the value of the other factors of production. Also, the marginal product of labor and capital will be positive in case of constant returns to scale.

The constant elasticity of substitution production function can be expressed algebraically as:

$$\boldsymbol{Q = A[\alpha K^{-\beta} + (1-\alpha)L^{-\beta}]^{-1/\beta}}$$

Where, Q = output, K = Capital and L = Labor

A = efficiency parameter that shows the organizational aspects of production and the state of technology.

The Constant elasticity of substitution production function shows, that any change in the technology or organizational aspects, the production function changes with a shift in the efficiency parameter.

α= distribution parameter or capital intensity factor coefficient concerned with relative factors in the total output.

β = substitution parameter, that determines the elasticity of substitution

The homogeneity of the production function can be determined by the value of the substitution parameter (β), if it is equal to one, then it is said to be linearly homogeneous i.e. the proportionate change in the input factors results in the increase in the output in the same proportion.

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**For example:**

Consider the following production function.

$$\mathbf{Q = 5K + 10L}$$

It means 1 unit K produces 5 units and 2 units of L produce 10 units.

If we assume that, initially K = 1, L = 2, the total output may be obtained by substituting 1 for K and 2 for L in the below equation. Thus,

$$\mathbf{Q_{x1} = 5(1) + 10(2)}$$

$$\mathbf{25 = 5 + 20}$$

when inputs are doubled (i.e., K =2 and L = 4). Then,

$$\mathbf{Q_{x2} = 5(1x2) + 10(2x2)}$$

$$\mathbf{50 = 10 + 40}$$

Thus, the given production functions, when inputs are doubled, the output is also doubled.

# TYPES OF COSTS

Profits are the difference between selling price and cost of production. In general the selling price is not within the control of a firm but many costs are under its control. The firm should therefore aim at controlling and minimizing cost. The various relevant concepts of cost are:

**1. Opportunity costs and Outlay costs:**

Opportunity costs refer to the 'costs of the next best alternative foregone'. We have scarce resources and all these have alternative uses. Where there is an alternative, there is an opportunity to reinvest the resources. In other words, if there are no alternatives, there are no opportunity costs. It is necessary that we should always consider the cost of the next best alternative foregone before committing the funds on a given option. In other words, the benefits from the present option should be more than the benefits of the next best alternative. Opportunity cost is said to exist when the resources are scarce and there are alternative uses for these resources. If there is no alternative, Opportunity cost is zero.

For example: if a firm owns a land, there is no cost of using the land (i.e., the rent) in firm's account. Bust the firm has an opportunity cost of using this land, which is equal to the rent foregone by not letting the land out on (the return of second best alternative is regarded as the cost of first best alternative) rent.

Out lay costs are as actual costs which are actually incurred by the firm. these are the payments made for labour, material, plant, building, machinery traveling, transporting etc., These are all those expenses appearing in the books of account, hence based on accounting cost concept.

**2. Explicit costs and Implicit costs:**

Explicit costs are also called as out-of-pocket cost that involve cash payments. These are the actual or business costs that appear in the books of accounts. These costs include payment of wages and salaries, payment for raw-materials, interest on borrowed capital funds, rent on hired land, Taxes paid etc.

Implicit costs are also called as imputed costs which don't involve payment of cash as they are not actually incurred. They would have been incurred had the owner not been in possession of the facilities. Ex: Interest on own capital, saving in terms of salary due to own supervision and rent of own building etc.

**3. Historical costs and Replacement costs:**

Historical cost is the original cost that has been originally spent to acquire the asset. of an asset. Historical valuation is the basis for financial accounts.

A replacement cost is the price that is to be paid currently to replace the same asset. A replacement cost is a relevant cost concept when financial statements have to be adjusted for inflation.

**4. Short – run costs and Long – run costs:**

Short-run is a period during which the physical capacity of the firm remains fixed. Any increase in output during this period is possible only by using the existing physical capacity more extensively. So short run cost is that which varies with output when the plant and capital equipment are constant.

Long run is defined as a period of adequate length during which a company may alter all factors of production with high degree of flexibility.

**5. Out-of pocket costs and Books costs:**

Out-of pocket costs also known as explicit costs are those costs that involve current cash payment such as purchase of raw material, payment of salary rents payment, interest on loan etc.

Book costs also called implicit costs do not require current cash payments. Depreciation, unpaid interest, salary of the owner is examples of back costs.

**6. Fixed costs, Variable costs and Semi-variable costs:**

Fixed cost is that cost which remains constant for a certain level to output. It is not affected by the changes in the volume of production but fixed cost per unit decreases, when the production is increased. Fixed cost includes salaries, Rent, Administrative expenses depreciations etc.

Variable is that which varies directly with the variation in output. An increase in total output results in an increase in total variable costs and decrease in total output results in a proportionate deccease in the total variables costs. The variable cost per unit will be constant. Ex: Raw materials, labour, direct expenses, etc.

Semi-variable costs refer to such costs that are fixed to some extent beyond which they are variable. Ex: telephone charges, Electricity charges, etc.

**7. Past costs and Future costs:**

Past costs also called historical costs are the actual cost incurred and recorded in the book of account these costs are useful only for valuation and not for decision making.

Future costs are costs that are expected to be incurred in the futures. They are not actual costs. They are the costs forecasted or estimated with rational methods. Future cost estimate is useful for decision making because decision are meant for future.

**8. Separable costs and Joint costs:**

The costs which can be traced or identified directly with a particular unit, department, or a process of production are called separable costs or direct costs or traceable costs.

The costs which cannot be identified directly with a particular unit, department or a process of the production are called joint costs or indirect costs or common costs. These costs are apportioned among various departments. Ex: Rent, Electricity, Administration salaries, Research and Development expenses etc.

**9. Avoidable costs and Unavoidable costs:**

Avoidable costs are those costs, which can be reduced if the business activities of a concern are curtailed. For example, if some workers can be retrenched with a drop in production, the wages of the retrenched workers are escapable costs.

Unavoidable costs are those that are essential for the sustenance of the business activity and hence they have to be incurred.

**10. Controllable costs and Uncontrollable costs:**

Controllable costs are ones, which can be regulated by the executive who is in charge of it. Direct expenses like material, labour etc. are controllable costs.

Some costs are not directly identifiable with a process of product. They are apportioned to various processes or products in some proportion. These apportioned costs are called uncontrollable costs.

**11. Incremental costs and Sunk costs:**

Incremental cost also known as differential cost is the additional cost due to a change in the level or nature of business activity. The change may be caused by adding a new product, adding new machinery, replacing a machine by a better one etc.

Sunk costs are those which are not altered by any change – They are the costs incurred in the past. This cost is the result of past decision, and cannot be changed by future decisions. Investments in fixed assets are examples of sunk costs. Once an asset is bought, the funds are blocked forever. They can neither be changed nor controlled.

**12. Total costs, Average costs and Marginal costs:**

Total cost is the total expenditure incurred for the input needed for production. It may be explicit or implicit. It is the sum total of the fixed and variable costs.

Average cost is the cost per unit of output. It is obtained by dividing the total cost (TC) by the total quantity produced (Q)

Marginal cost is the additional cost incurred to produce an additional unit of output.

**13. Accounting costs and Economic costs:**

Accounting costs are the costs recorded for the purpose of preparing the profit & loss account and balance sheet to meet the legal, financial and tax purpose of the company. The accounting concept is a historical concept and records what has happened in the post.

Economic cost refers to cost of economic resources used in production including opportunity cost. Economics concept considers future costs and future revenues, which help future planning, and choice, while the accountant describes what has happened, the economics aims at projecting what will happen.

14. **Urgent costs and Postponable costs:**

Urgent cost are those costs such as raw materials, wages and son, necessary to sustain the productivity.

Postponable costs are those costs which can be conveniently postponed. For example, white washing the building etc.

## COST-OUTPUT RELATIONSHIP

The cost-output relationship plays an important role in determining the optimum level of production. Knowledge of the cost-output relationship helps the manager in cost control, profit prediction, pricing, promotion etc. Output is an important factor, which influences the cost. Considering the period the cost function can be classified as (a) short-run cost function and (b) long-run cost function. In the short run, the costs can be classified into fixed costs and variable costs. The cost-output relationship in the short run is governed by certain restrictions in terms of fixed costs whereas in the long run, the cost-output relationship studies the effect of varying the size of plants upon its cost.

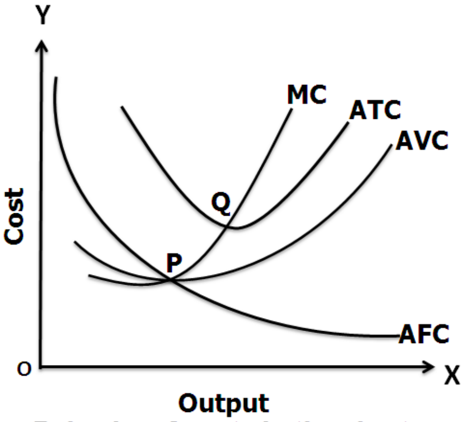
### I. Short-run Cost Function:

Under short run cost-output relation, costs in short run are classified into fixed costs and variable costs. Labour is the variable factor while capital is the fixed factor. Total fixed cost remains constant while variable cost changes with the variation in units if labour. The fixed costs may be ascertained in terms of total fixed cost and average fixed cost per unit. The variable cost can be determined in terms of average variable cost, total variable cost. The below table explains the behavior of costs in the short run. From the below it is clear that:

**Cost – Output relationship**

| Output(Q) (Units) | Total fixed cost TFC (Rs) | Total variable cost TVC (Rs) | Total cost (TFC + TVC) TC (Rs) | Average variable cost (TVC / Q) AVC (Rs) | Average fixed cost (TFC / Q) AFC (Rs) | Average Total cost (TC/Q) AC (Rs) | Marginal cost MC (Rs) |
|---|---|---|---|---|---|---|---|
| 0 | - | - | 60 | - | - | - | - |
| 1 | 60 | 20 | 80 | 20 | 60 | 80 | 20 |
| 2 | 60 | 36 | 96 | 18 | 30 | 48 | 16 |
| 3 | 60 | 48 | 108 | 16 | 20 | 36 | 12 |
| 4 | 60 | 64 | 124 | 16 | 15 | 31 | 16 |
| 5 | 60 | 90 | 150 | 18 | 12 | 30 | 26 |
| 6 | 60 | 132 | 192 | 22 | 10 | 32 | 42 |

- Total fixed costs remain fixed irrespective of increase or decrease in production activity.
- Average fixed cost per unit declines as the volume of production increases. As production increases, the fixed costs are spread over a great number of units. There is inverse relationship between average fixed cost and volume of production.
- The total variable cost increases proportionately with production. But, the rate of increase is not constant.
- The total cost increases with the volume of production.
- The average total cost decreases up to certain level of production. After this level, it rises steeply. It results in flat U-shaped curve. The lowest point of AFC curve denotes the ideal level of production.
- Marginal cost is the change in total cost due to one unit change in output.




**Behavior of costs in the short run**

The short-run cost-output relationship can be shown graphically as follows.

- From this graph, it is noticed that as the production increases, the AFC will continue to decrease. Hence, the AFC curve will slope downwards and it appears to meet the X axis but it never meets the X axis.
- The AVC curve is U-shaped curve indicating that the AVC curve tends to fall in the beginning when output increases but after a particular level of output, it rises because of the application of law of returns or law of variable proportions or law of diminishing returns.
- The ATC curve initially declines and then rises upward. As long as AVC declines the ATC will also decline.
- MC is the change in total cost resulting from a unit change in output. It decreases up to certain level of output but later rises steeply.

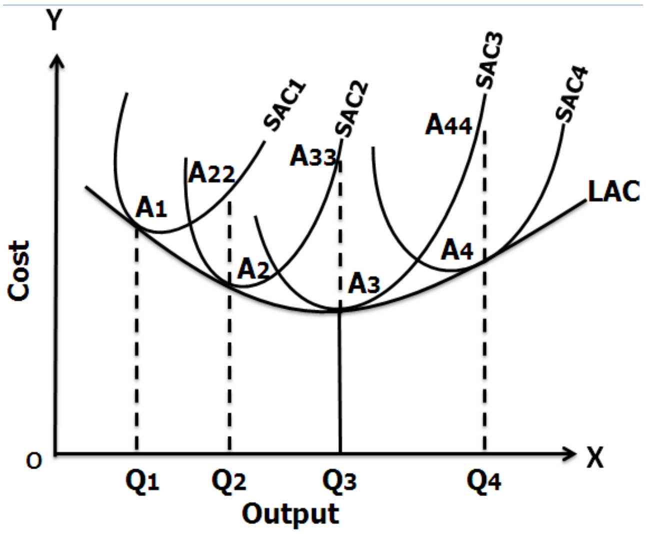
## II. Long-run Cost Function:

Long run refers to that period of time over which all factors are variable. It has no fixed cost. Over a long period, the size of the plant can be changed, unwanted buildings can be sold staff can be increased or reduced. The long run enables the firms to expand and scale of their operation by bringing or purchasing larger quantities of all the inputs. Thus in the long run all factors become variable.

In the long run a firm has a number of alternatives in regards to the scale of operations. For each scale of production or plant size, the firm has an appropriate short-run average cost curves. The short-run average cost (SAC) curve applies to only one plant whereas the long-run average cost (LAC) curve takes in to consideration many plants.

If we assume that there are many plant sizes, each suitable for a certain level of output, we will get many SAC curves intersecting each other. As the number of plant sizes increases, the points of intersection of SAC curve will come closer. And, if we assume that there are large number (say, infinite number) of plant sizes the intersection points will be so near to each other that we get almost a continuous curve. Thus continuous curve is known as the Long-run Average Cost (LAC) curve or the Envelope curve (as it envelopes the family of short-run Average Cost Curves).

The long-run cost-output relationship is shown graphically with the help of "LCA' curve.




The above figure shows how LAC curve envelopes several short-run average cost (SAC) curves. Suppose, the firm is producing an output of OQ1 units on a plant of SAC1, if it wants to produce O Q2 units of output, either it can operate on SAC1 by over utilizing SAC1 plant or by acquiring a bigger size paIntSAC2 and operate on it. It will be less costly to operate on SAC2. If it wants to produce O Q3 units of output, it can operate on the bigger size plant SAC3 at least cost. Q3A3 is the least cost at the output OQ3 and the firm attains optimum output in the long-run at OQ3 level of output. If it operates on SAC2 to produce OQ3 units of output, the cost will be prohibitively high being Q3A33. It is to be notes that there is only one short-run average cost curve SAC3 which is tangential to the long-run average cost curve at its minimum point. All other SAC curves are tangential to the LAC curves at higher than their minimum average cost points.

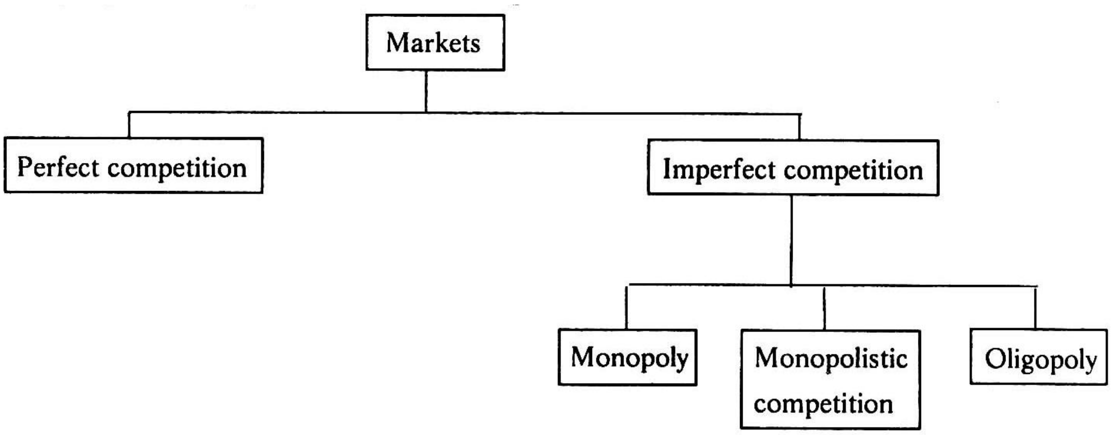
## MARKET

Market is a place where buyer and seller meet, goods and services are offered for the sale and transfer of ownership occurs. A market may be also defined as the demand made by a certain group of potential buyers for a good or service. The former one is a narrow concept and later one is a broader concept. Economists describe a market as a collection of buyers and sellers who transact over a particular product or product class (the housing market, the clothing market, the grain market etc.). For business purpose we define a market as people or organizations with wants (needs) to satisfy, money to spend, and the willingness to spend it. Broadly, market represents the structure and nature of buyers and sellers for a commodity/service and the process by which the price of the commodity or service is established. In this sense, we are referring to the structure of competition and the process of price determination for a commodity or service. The determination of price for a commodity or service depends upon the structure of the market for that commodity or service (i.e., competitive structure of the market). Hence the understanding on the market structure and the nature of competition are a pre-requisite in price determination.

**Different Market Structures**

Market structure describes the competitive environment in the market for any good or service. A market consists of all firms and individuals who are willing and able to buy or sell a particular product. This includes firms and individuals currently engaged in buying and selling a particular product, as well as potential entrants.

The determination of price is affected by the competitive structure of the market. This is because the firm operates in a market and not in isolation. In making decisions concerning economic variables it is affected, as are all institutions in society by its environment.



## PERFECT COMPETITION

Perfect competition refers to a market structure where competition among the sellers and buyers prevails in its most perfect form. In a perfectly competitive market, a single market price prevails for the commodity, which is determined by the forces of total demand and total supply in the market.

**Characteristics Of Perfect Competition**

The following features characterize a perfectly competitive market:

a) **A large number of buyers and sellers:** The number of buyers and sellers is large and the share of each one of them in the market is so small that none has any influence on the market price.

b) **Homogeneous product:** The product of each seller is totally undifferentiated from those of the others.

c) **Free entry and exit**: Any buyer and seller is free to enter or leave the market of the commodity.

d) **Perfect knowledge**: All buyers and sellers have perfect knowledge about the market for the commodity.

e) **Indifference:** No buyer has a preference to buy from a particular seller and no seller to sell to a particular buyer.

f) **Non-existence of transport costs**: Perfectly competitive market also assumes the non-existence of transport costs.

g) **Perfect mobility of factors of production**: Factors of production must be in a position to move freely into or out of industry and from one firm to the other.

**Perfect competition: The individual firm**

AR(Average revenue) curve and MR(Marginal Revenue) curve under perfect competition becomes equal to D(Demand) curve and it would be a horizontal line or parallel to the X-axis. The curve simply implies that a firm under perfect competition can sell as much quantity as it likes at the given price determined by the industry i.e. a perfectly elastic demand curve.

**Price=AR=MR=D**

| Quantity Q | Price P | Total Revenue TR | Average Revenue AR | Marginal Revenue MR |
|---|---|---|---|---|
| 1 | 5 | 5 | 5 | 5 |
| 2 | 5 | 10 | 5 | 5 |
| 3 | 5 | 15 | 5 | 5 |
| 4 | 5 | 20 | 5 | 5 |


**Fig: Demand curve for the firm in perfect competition**

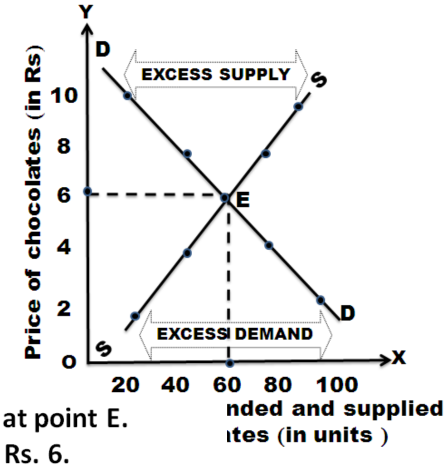
**Perfect competition: The firm and the industry**

Price is determined by the market forces, that is, demand and supply for a given product or service. As discussed above, firms have no control over the prices they charge for their products. The ultimate price that determines the quantity demanded is equal to the quantity supplied. This price is also called equilibrium price, as it balances the forces of demand and supply. The figure shows how the price is determines. DD is the demand curve and SS is the supply curve. Rs. 6 is the price at which DD and SS intersect each other. At Rs. 6, 60 units are supplied and demanded.

If the price increases to Rs.8, supply will also increase and hence the price is likely to fall down.

If the price decreases to Rs. 4, supply will decrease and hence the price is likely to go up.

| Price P | Qty. Demanded D | Qty. Supplied S | |
|---|---|---|---|
| 2 | 100 | 20 | |
| 4 | 40 | 80 | |
| 6 | 60 | 60 | Equilibrium state |
| 8 | 40 | 80 | |
| 10 | 20 | 100 | |




- ➢Market equilibrium is determined at point E.
- ➢Equilibrium price is determined at Rs. 6.
- ➢Equilibrium quantity is determined at 60 chocolates.

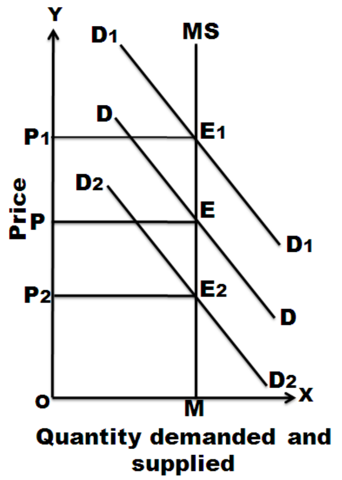
**Price-Output Determination Under Perfect Competition**

In this market, the price is determined by supply and demand forces. Marshal who propounded the theory says that the price is determined by the equilibrium between demand and supply.

The pricing of commodity under perfect competition can be determined in three periods of time.

**a) Very short period** (Market Period)

Market period is too short period to increase the supply. The market period is so short that supply of the commodity is limited to existing stock. During the market period, say a single day, the supply of a commodity is perfectly inelastic.

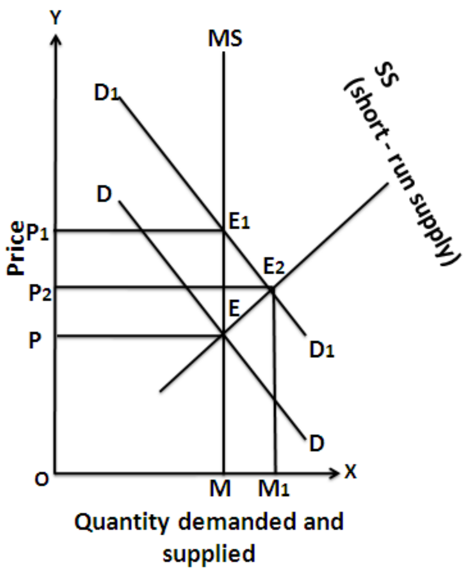



In this figure quantity is represented along X-axis and price is represented along Y-axis. MS is the very short period supply curve. DD is demand curve. It intersects supply curve at E. The price is OP. The quantity is OM. D1 D1 represents increased demand. This curve cuts the supply curve at E1. Even at the new equilibrium, supply is OM only. But price increases to OP1. So, when demand increases, the price will increase but not the supply. If demand decreases new demand curve will be D2 D2. This curve cuts the supply curve at E2. Even at this new equilibrium, the supply is OM only. But price falls to OP2. Hence in very short period, given the supply, it is the change in demand that influences price. The price determined in a very short period is called Market Price.

**b) Short Period**

Short period is not too long period to install new capital equipments. It is also not sufficient period to permit the new firms to enter the industry to increase the supply of the commodity in the market. Hence the firm can increase the supply of a commodity in the short period only by making intensive use of the given plants and equipments and increasing the units of variable factors.

As a result of this, the short period supply of a commodity will be relatively less elastic.

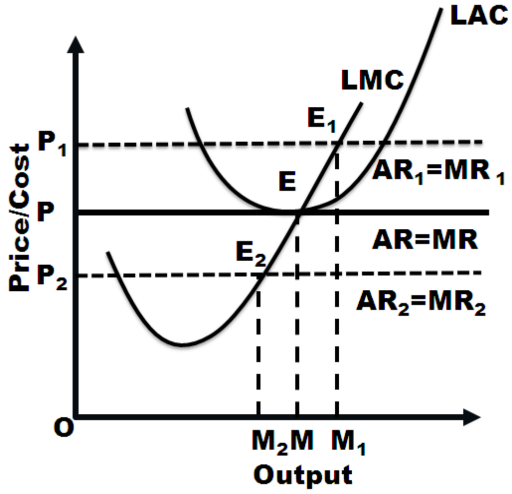


In the diagram MS is the market period supply curve. DD is the initial demand curve. It intersects MS curve at E. The price is OP and output OM. Suppose demand increases, the demand curve shifts upwards and becomes D1D1. In the very short period, supply remains fixed on OM. The new demand curve D1D1 intersects MS at E1. The price will rise to OP1. This is what happened in the very short-period.

As the price rises from OP to OP1, firms expand output. As firms can vary some factors but not all, the law of variable proportions operates. This results in new short-run supply curve SS. It interests D1 D1 curve at E2. The price will fall from OP1 to OP2.

**c) Long Period**

In Long run, the Firm's output (supply) can be changed by both the variable factors and fixed factors i.e. all factors become variable. There is enough time for new Firms to enter the Industry. Further, if the demand is increased, the supply can be increased or decreased according to the demand. For Long run equilibrium, long run marginal cost (LMC) is equal to MR and LMC curve cut the MR curve from below. In case of long run equilibrium, all the firms will earn only normal profits.




Take the case when the Firm earn super-normal profit-Then the existing Firm will increase their production and new Firm will enter the Industry. Consequently, the total supply will increase and price fall down and further results in normal profit for the firm

On the contrary, if the firm is incurring losses, Then some Firm will leave the Industry which will reduce the total supply. And due to decrease in supply, price will rise and once again Firm will begin to earn normal profit. Firm equilibrium is at the minimum point of its LAC and at this point the Firm will get the normal profits. If AR (price) rises to $OP_1$, then Firm's LMC cuts its MR1 at $E_1$ and the firm gets super-normal profit but again come to OP yielding normal profits as stated before. And at price $OP_{2,}$ firm incurs losses but again rise to level OP to maintain the equilibrium at normal profit

Firm's equilibrium: MC=MR=AR= min LAC

## MONOPOLY

'mono' means single and 'poly' means seller. The term monopoly refers to that market in which a single firm controls the whole supply of a particular product which has no close substitutes. Monopoly emerges in firms such as transport, water and electricity supply etc.

**Features:**

1. **Single person or a firm:** A single person or a firm controls the total supply of the commodity. There will be no competition for monopoly firm. The monopolist firm is the only firm in the whole industry.
2. **No close substitute:** The goods sold by the monopolist shall not have close substitutes. Even if price of monopoly product increases, people will not go in far substitute. For example: If the price of electric bulb increases slightly, consumer will not go in for kerosene lamp.
3. **Large number of Buyers:** Under monopoly, there may be a large number of buyers in the market who compete among themselves.
4. **Price Maker:** Since the monopolist controls the whole supply of a commodity, he is a price-maker, and then he can alter the price.

5. **Supply and Price:** The monopolist can fix either the supply or the price. He cannot fix both. If he charges a very high price, he can sell a small amount. If he wants to sell more, he has to charge a low price. He cannot sell as much as he wishes for any price he pleases.
6. **Downward Sloping Demand Curve:** The demand curve (average revenue curve) of monopolist slopes downward from left to right. It means that he can sell more only by lowering price.

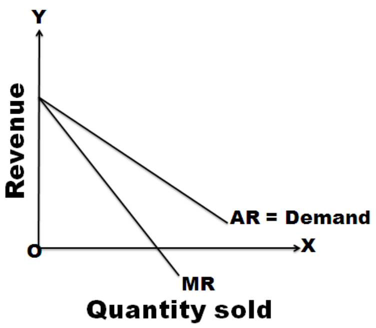
Monopoly refers to a market situation where there is only one seller. He has complete control over the supply of a commodity. He is therefore in a position to fix any price. Under monopoly there is no distinction between a firm and an industry. This is because the entire industry consists of a single firm.

Being the sole producer, the monopolist has complete control over the supply of the commodity. He has also the power to influence the market price. He can raise the price by reducing his output and lower the price by increasing his output. Thus he is a price-maker. He can fix the price to his maximum advantages. But he cannot fix both the supply and the price, simultaneously. He can do one thing at a time. If he fixes the price, his output will be determined by the market demand for his commodity. On the other hand, if he fixes the output to be sold, its market will determine the price for the commodity. Thus his decision to fix either the price or the output is determined by the market demand.

The market demand curve of the monopolist (the average revenue curve) is downward sloping. Its corresponding marginal revenue curve is also downward sloping. But the marginal revenue curve lies below the average revenue curve as shown in the figure. The monopolist faces the down-sloping demand curve because to sell more output, he must reduce the price of his product. The firm's demand curve and industry's demand curve are one and the same. The average cost and marginal cost curve are U shaped curve. Marginal cost falls and rises steeply when compared to average cost.

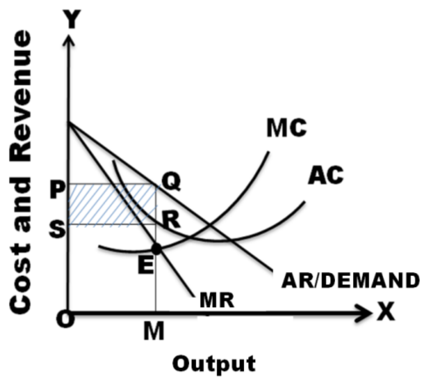
Under monopoly, demand curve is average revenue curve.

| Qty | AR/Price | TR | MR |
|---|---|---|---|
| 1 | 8 | 8 | 8 |
| 2 | 6 | 12 | 4 |
| 3 | 4 | 12 | 0 |
| 4 | 2 | 8 | -4 |
| 5 | 0 | 0 | -8 |




**Price-Output Determination Under Monopoly**

The monopolistic firm attains equilibrium when its marginal cost becomes equal to the marginal revenue. The monopolist always desires to make maximum profits. He makes maximum profits when MC=MR. He does not increasing his output if his revenue exceeds his costs. But when the costs exceed the revenue, the monopolist firm incur loses. Hence the monopolist curtails his production. He produces up to that point where marginal cost is equal to the marginal revenue (MR=MC). Thus, the point is called equilibrium point. The price output determination under monopoly may be explained with the help of a diagram.



In the diagram, the quantity supplied or demanded is shown along X-axis. The cost or revenue is shown along Y-axis. AC and MC are the average cost and marginal cost curves respectively. AR and MR curves slope downwards from left to right. AC and MC are U shaped curves. The monopolistic firm attains equilibrium when its marginal cost is equal to marginal revenue (MC=MR). Under monopoly, the MC curve may cut the MR curve from below or from a side. In the diagram, the above condition is satisfied at point E. At point E, MC=MR. The firm is in equilibrium. The equilibrium output is OM. Up to OM output, MR is greater than MC and beyond OM, MR is less than MC. Therefore, the monopolist is will be in equilibrium at output OM where MR=MC and profits are maximized.

The above diagram (Average revenue) = MQ or OP

Average cost = MR

Profit per unit = Average Revenue-Average cost=MQ-MR=QR

Total Profit = QR x SR=PQRS

If AR > AC;  Abnormal or super normal profits.

If AR = AC;  Normal Profit

If AR < AC ;   Loss

## MONOPOLISTIC COMPETITION

Perfect competition and pure monopoly are rare phenomena in the real world. Instead, almost every market seems to exhibit characteristics of both perfect competition and monopoly. Hence, in the real world, it is the state of imperfect competition lying between these two extreme limits that work. Edward. H. Chamberlain developed the theory of monopolistic competition, which presents a more realistic picture of the actual market structure and the nature of competition.

**Features/Characteristics**

The important characteristics of monopolistic competition are:

1. **Existence of Many firms:** Industry consists of a large number of sellers, each one of whom does not feel dependent upon others. Every firm acts independently without bothering about the reactions of its rivals. The size is so large that an individual firm has only a relatively small part in the total market, so that each firm has very limited control over the price of the product. As the number is relatively large, it is difficult for these firms to determine its price- output policies without considering the possible reactions of the rival forms. A monopolistically competitive firm follows an independent price policy.
2. **Product Differentiation:** product differentiation is the essential feature of monopolistic competition. Products can be differentiated by means of unique facilities, advertising, brand loyalty, packing, pricing, terms of credit, superior maintenance service, convenient location and so on. Through heavy advertisement budgets, Pepsi and Coca-Cola make it very expensive for a third competitor to enter the cola market on such a big scale. The following example illustrate how the firms differentiate themselves from others in a monopolistic environment.
    - In hotel industry, some hotels have spacious swimming pools, gyms, cultural programs etc. The customers who value these facilities don’t bother about price changes.
    - The colleges who provide best infrastructure and placements in various reputed companies have demand from the student community irrespective of an increase in tuition fee.
    - Cell phones which have unique features have demand from the public even price increases.

3. **Large Number of Buyers:** There are large number of buyers in the market. But the buyers have their own brand preferences. So, the sellers are able to exercise a certain degree of monopoly over them. Each seller has to plan various incentive schemes to retain the customers who patronize his products.
4. **Free Entry and Exist of Firms:** As in the perfect competition, in the monopolistic competition too, there is freedom of entry and exit. That is, there is no barrier as found under monopoly.
5. **Selling costs:** Since the products are close substitutes, much effort is needed to retain the existing consumers and to create new demand. So, each firm has to spend a lot on selling cost, which includes cost on advertising and other sale promotion activities.
6. **Imperfect Knowledge:** Imperfect knowledge about the product leads to monopolistic competition. If the buyers are fully aware of the quality of the product, they cannot be influenced much by advertisement or other sales promotion techniques.
7. **The Group:** Under perfect competition, the term industry refers to all collection of firms producing a homogenous product. But under monopolistic competition, the products of various firms are not identical though they are close substitutes.

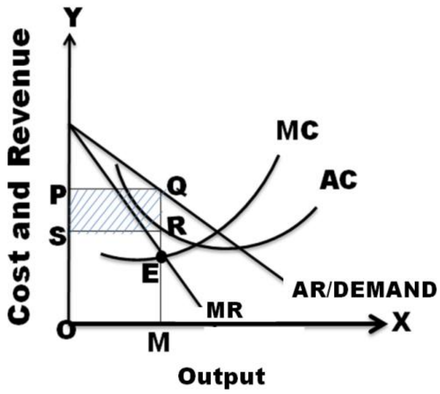
**Price – Output Determination Under Monopolistic Competition**

Under monopolistic competition, Since different firms produce different varieties of products, different prices for them will be determined in the market depending upon the demand and cost conditions. Each firm will set the price and output of its own product. Here also the profit will be maximized when marginal revenue is equal to marginal cost(MR=MC). The demand curve for the firm in case of monopolistic competition is just similar to that of monopoly.

The degree of elasticity of demand of a firm in monopolistic competition depends upon the extent to which the firm can resorts to product differentiation. The greater the ability of the firm to differentiate the product, the less elastic the demand is. The firm’s influence to increase the price depends upon the extent to which it can differentiate the product.

a) **Short-run**

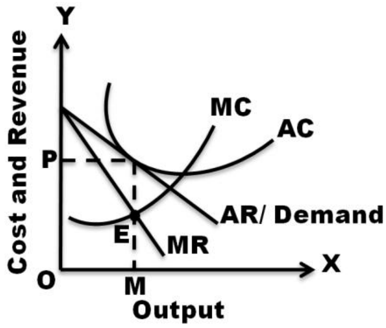
In the short-run, the firm is in equilibrium when marginal Revenue = Marginal Cost. In the figure, AR is the average revenue curve. MR marginal revenue curve, MC marginal cost curve, AC average cost curve, MR and MC interest at point E where output is OM and price MQ (i.e. OP). Thus, the equilibrium output is OM and the price is MQ or OP. When the price (average revenue) is above average cost, a firm will be making supernormal profit. From the figure it can be seen that AR is above AC in the equilibrium point. As AR is above AC, this firm is making abnormal profits in the short-run. The abnormal profit per unit is QR, i.e., the




difference between AR and AC at equilibrium point and the total supernormal profit is OR x OM. This total abnormal profits is represented by the rectangle PQRS. The firm may make supernormal profits in the short-run if it satisfies the following two conditions.

a) MR = MC
b) AR > AC

**b) Long–run**




More and more firms will be entering the market having been attracted by supernormal profits enjoyed by the existing firms in the industry. As a result, competition become s intensive on one hand, forms will compete with one another for acquiring scare inputs pushing up the prices of factor inputs. On the other hand, on the entry of several firms, the supply in the market will increase, pulling down the selling price of the products. In order to cope with the competition, the firms will have to increase the budget on advertising. The entry of new firms continue till the supernormal profits of the firms completely eroded and ultimately firms in the industry will earn only normal profits. Those firms which are not able to earn at least normal profits will get closed. Thus in the long-run, every firm in the monopolistic competitive industry will earn only normal profits, which are just sufficient to stay in the business. It is be noted that normal profits are part of average costs.

In the long-run, in order to achieve equilibrium position, the firm has to fulfill the following conditions:

a) MR = MC

b) AR = AC at the level of equilibrium level of output.

## OLIGOPOLY

The term oligopoly is derived from two Greek words, oligos meaning a few, and pollen meaning to sell. Oligopoly is the form of imperfect competition where there are a **few sellers** in the market, producing either a homogeneous product or producing products, which are close but not perfect substitute of each other.

**Features**

1. **Monopoly Power:**

   There is a clement of monopoly power in oligopoly. Since there are only a few firms and each firm has a large share of the market. In its share of the market, it controls the price and output. Thus an oligopoly has some monopoly power.

2. **Interdependence of Firms:**

   Under oligopoly, there are only a few firms, each producing a homogeneous or slightly differentiated product. Since the number of firms is small, each firm enjoys a large share of the market and has a significant influence on the price and output decisions. Thus, there is interdependence of firms. No firm can ignore the actions and reactions of rival firms under oligopoly.

3. **Conflicting Attitude of Firms:**

   Under oligopoly, two types of conflicting attitudes are found in the firms. On the one hand, firms realize the disadvantages of mutual competition and desire to combine to maximize their joint profits. This tendency leads to the formation of collusion. On the other hand, the desire to maximize one's individual profit may lead to conflict and antagonism; the firms come into clash with one another on the question of distribution of profits and allocation of markets. Thus, there is an existence of two opposing attitudes among the firms.

4. **Few firms. In this market, only few sellers are found:**

   For example, the market for automobiles in India exhibits oligopolistic structure as there are only few producers of automobiles. If there are only two firms, it is called 'duopoly'.

5. **Nature of product:**

   If the firms product homogeneous product, it becomes pure oligopoly. The firms with product differentiation constitute impure oligopoly.

6. **Interdependence among firms:**

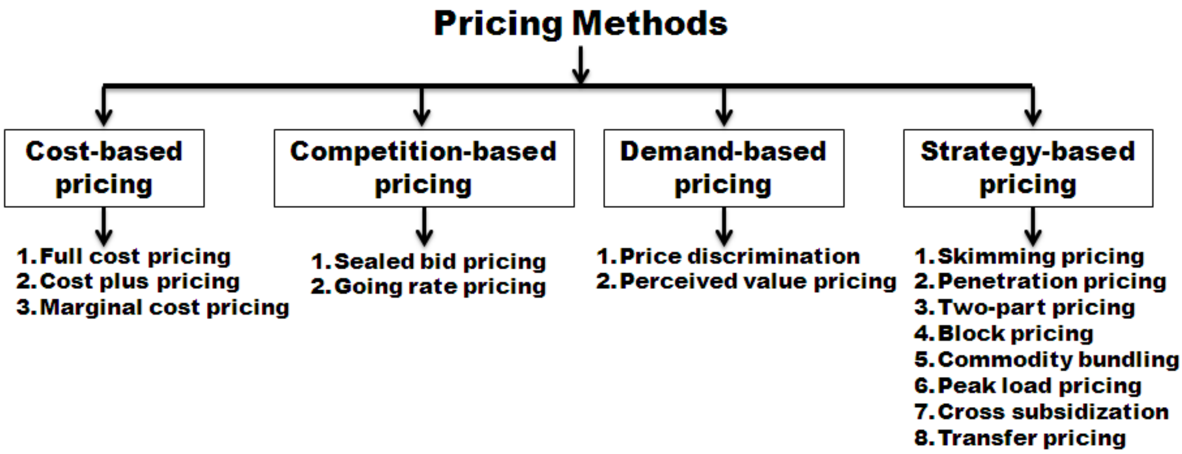
   In oligopoly market, each firm treats the other as its rival firm. It is for this reason that each firm while determining price of its product, takes into account the reaction of the other firms to its own action.

7. **Large number of consumers:**

   In this market, there are large numbers of consumers to demand the product.

## TYPES OF PRICING

Firms set prices for their products through several alternative means. The important pricing methods followed in practice are shown in the chart.




### A. Cost Based Pricing

1. **Full cost pricing:-**Under this method, price is just equal to the average cost.
2. **Cost plus pricing:-** Here, the average cost is ascertained and then a conventional margin added to the cost to arrive at the price. In other words, find out the product unit's total cost and add a percentage of profit to arrive at the selling price. It is commonly followed in departmental stores and other retail shops. This method is simple to be administered. It may be very difficult to find the selling price in advance due to complexity of the nature of the project.
3. **Marginal cost pricing(Break even pricing or Target profit pricing):-** In this method, selling price is fixed in such a way that it covers full variable or marginal cost and contributes towards recovery of fixed costs. in the stiff competition, marginal cost offers a guidelines as to how far the selling price can be lowered.

### B. Competition based pricing

Here the price of product is set based on what the competitor charges for a similar product. In other words, a reduction in the price of products by the competitor will force us to follow suit. In such a case, how far we can go on reducing the price?. Here the marginal cost concept comes handy. As long as the price covers the marginal cost, continue to sell. If not, better stop selling. It is because, every unit sold at less than marginal cost results in loss.

1. **Sealed bid pricing**:- This method is more popular in tenders and contracts. Each contracting firm quotes its price in a sealed cover called "tender". All the tenders are opened on a scheduled date and the person who quotes the lowest price is awarded the contract.
2. **Going rate pricing**:- Here the prevailing market price is charged. Suppose, when one wants to buy or sell gold, the prevailing market rate at a given point of time is taken as the basis to determine the price.

**C. Demand Based Pricing**

1. **Perceived value pricing**:- This method considers the buyer's perception of the value of the product as the basis of pricing. Here the pricing rule is that the firm must develop procedures for measuring the relative value of the product as perceived by consumers.
2. **Price discrimination(Differential pricing):-** Price discrimination refers to the practice of charging different prices to customers for the same good. In involves selling a product or service for different prices in different market segments. Price differentiation depends on geographical location of the consumers, type of consumer, purchasing quantity, season, time of the service etc. E.g. Telephone charges, APSRTC charges.

**D. Strategy based pricing**

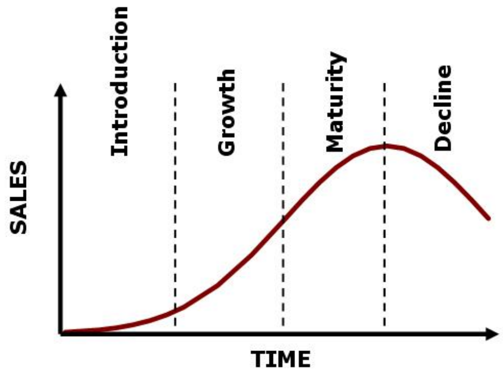
1. **Skimming pricing:-** The company follows this method when the product is for the first time introduced in the market. Under this method, the company fixes a very high price for the product. this strategy is mostly found in case of technology products. When Samsung introduces a new cell phone model, it fixes a high price because of the uniqueness of the product.
2. **Penetration pricing:-** This is exactly opposite to the market skimming method. Here, a low price is fixed for the product in order to catch the attention of consumers, once the product image and credibility is established, the seller slowly starts jacking up the price to reap good profits in future. The Rin washing soap perhaps falls into this category. This soap was sold at a rather low price in the beginning and the firm even distributed free samples. Today, it is quite an expensive brand and yet it is selling very well.
3. **Two-part pricing:-** Under this strategy, a firm charges a fixed fee for the right to purchase its goods, plus a per unit charge for each unit purchased. Entertainment houses such as country clubs, athletic clubs, etc, usually adopt this strategy. They charge a fixed initiation fee or membership fee plus a charge, per month or per visit, to use the facilities.
4. **Block pricing:-** We see block pricing in our day-to-day life very frequently. Four Santhoor soaps in a single pack with nice looking soap box or five Maggi packets in a single pack with an attractive bowl indicate this pricing method. The total value of the goods includes consumer's surplus as the consumer is given soap box and bowl along with the products freely. By selling certain number of units of a product as one package, the firm earns more than by selling unit wise.
5. **Commodity bundling:-** Commodity bundling means the practice of bundling two or more different products together and selling them at single 'bundle price'. For example tourist companies offer the package that includes the travelling charges, hotel, meals and sight-seeing etc, at a bundle price instead of pricing each of these services separately.

6. **Peak load pricing:-** Under this method, high price is charged during the peak times than off-peak times. RTC increases charges during festivals, Railways charge more fares during tatkal time. During seasonal period when demand is likely to be higher, a firm may increase profits by peak load pricing.
7. **Cross subsidization:-** The process of charging high price for one group of customers in order to subsidize another group.
8. **Transfer pricing:-** Transfer pricing means a price at which one process forwards their output(work-in-progress) to the next process for further processing. It is an internal pricing technique.

## PRODUCT LIFE CYCLE BASED PRICING

Companies must adapt to the stages of the product life cycle to effectively sell and promote their products. Depending on the product life cycle stage, a company will develop branding techniques and an appropriate pricing model. Understanding each stage helps businesses increase profits.

The stages of a product life cycle govern how a product is priced, distributed, and promoted. A new product goes through multiple stages during the course of its life cycle, including an introduction stage, growth stage, maturity stage and a decline stage. As a product ages, companies look for new ways to brand it, and also explore pricing changes. Market and competitor research help businesses assess the proper course of action to maintain product profitability.




### Introduction Stage

A new product may simply be either another brand name added to the existing ones or an altogether new product. Pricing a new brand for which there are many substitutes available in market is not a big problem as pricing a new product for which close substitutes are not available.

There are two type of pricing strategies for new product.

1. ***Skimming price policy*:-** Selling a product at a high price, sacrificing high sales to gain a high profit, therefore 'skimming' the market. Usually employed to reimburse the cost of investment of the original research into the product - commonly used in electronic markets when a new range, such as DVD players, are firstly dispatched into the market at a high price.
2. ***Penetration price policy*:-** This pricing policy is adopted generally in the case of new product for which substitutes are available. This policy requires fixing a lower initial price designed to penetrate the market as quickly as possible.

### Growth Stage

During the growth stage, a company aims to develop brand recognition and increase their customer base. The quality of their product is often improved based on early reviews, and technical support is usually enhanced. Pricing remains generally stable as demand continues with minimal competition. A larger distribution network is formed to keep up with the pace of demand.

**Maturity Stage**

In the maturity stage, the steady sales start to decline and companies face greater challenges in the marketplace. Competitors will often introduce rival products with the intent of grabbing some of the market share. This is the product life cycle stage in which the customer base is heavily fought over and price decreases most often occur. Additional features are added to distinguish a product from its competitors. Companies run promotions during this stage that highlight the primary differences between their product and their competitor's products.

**Decline Stage**

In the decline stage, a company will make important decisions regarding the future of their product. They can choose to create new iterations of the product with new features, or they can reduce the price and offer it at a discount. A company may choose to discontinue the product altogether, either disposing of their inventory or selling it to another company who is willing to manufacture and market it. Promotion at this stage will depend on whether a company chooses to continue its product, and how they plan to re-market it.

## BREAK EVEN ANALYSIS

BEP analysis is also called as CVP analysis. The BEP can be defined as that level of sales at which total revenues equals total costs and the net income is equal to zero. This is also known as no-profit no-loss point.

Break-even analysis refers to analysis of costs and their possible impact on revenues and volume of the firm. Hence, it is also called the cost-volume-profit (CVP) analysis. A firm is said to attain the BEP when its total revenue is equal to total cost(TR=TC).

The main objective of the Break Even Analysis is not only to spot the BEP but also to develop an understanding of the relationships of cost, volume and price within a company's practical range of operations.

**Assumptions of Break-Even Analysis**

1. All cost are divided into fixed and variable
2. Fixed costs remain constant whereas variable costs vary
3. Selling price remains constant
4. There will be no change in the operating efficiency

**Key terms used in Break-even Analysis**

**1. Fixed cost(FC):-** Fixed cost remains fixed in the short-run. These costs must be borne by the firm ever there is no production. Example: Rent, Insurance, Depreciation, permanent employees' salaries. Etc. Fixed cost per units varies.

**2. Variable costs(VC)**:- The costs which vary in direct proportion to the production/sales volume are called as variable costs. variable cost per unit is fixed. Examples for variable costs: cost of direct material, cost direct labor, direct expenses, operating supplies such as oil, grease etc.

**3. Total cost(TC)**:- The total of fixed cost and variable costs. TC=FC+VC

**4. Total revenue:-** The sales amount of goods sold in the market.(Selling Price per unit x No of units sold).

**5. Contribution:-** The excess of sales revenue over variable cost(C=S-V).

**6. P/V Ratio(Profit/Volume Ratio):-** The ratio between the contribution and sales:

$$P/VRatio = \frac{\mathrm{C}}{\mathrm{S}} = \frac{\mathrm{S-V}}{\mathrm{S}} = \frac{\mathrm{F+P}}{\mathrm{S}} = 1 - \frac{\mathrm{V}}{\mathrm{S}} = \frac{\mathrm{P}}{\mathrm{M/S\ Sales}} = \frac{\mathrm{F}}{\mathrm{BEP\ Sales}} = \frac{\mathrm{Difference\ in\ profit/contribution}}{\mathrm{Difference\ in\ sales}}$$

**7. Margin of Safety sales(M/S sales):-** The excess of actual total sales over break even sales.

$$\mathrm{M/S\ sales} = \mathrm{Total\ sales} - \mathrm{Break\ even\ sales} = \frac{\mathrm{P}}{\mathrm{P/V\ Ratio}}$$

**8. Break-even point(BEP):-** The point where total revenue is just equal to the total cost is called Break-even point. At break-even point, there is not profit or no loss to the business. Break-even point can be calculated in units as well as in sales.

$$BEP(in\ units) = \frac{F}{C\ per\ unit}$$

$$BEP(in\ sales) = \frac{F}{P/VRatio}$$

$$Units\ for\ a\ desired\ profit = \frac{F+DP}{C\ per\ unit}$$

$$Sales\ for\ a\ desired\ profit = \frac{F+DP}{P/VRatio} = Units\ for\ desired\ profit\ x\ SP\ per\ unit$$

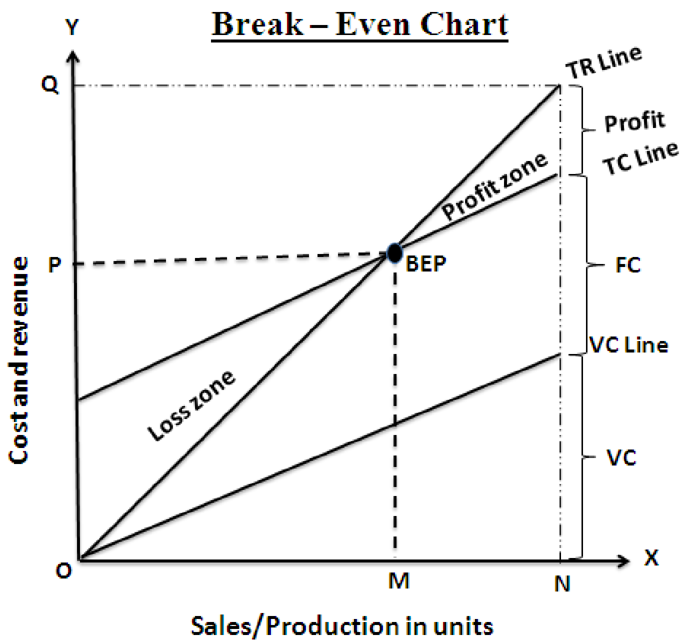
**S-V=F±P**
**S=F+V±P**
**P=S-(F+V)**

**C=Contribution**
**S=Sales**
**V=Variable cost**
**F=Fixed cost**
**P=Profit**
**BEP=Break Even Point**
**DP=Desired Profit**
**+P=Profit**
**-P=Loss**

**Important notes:**

- At BEP sales, Total sales=Total cost(F+V). At this stage, total contribution(S-V) is equal to fixed cost. So, there is no profit or no loss. Hence, C=F.
- Below the BEP sales, total contribution is not equal to total fixed cost. Hence, C<F.
- Beyond BEP sales(M/S sales), since fixed cost is recovered at BEP sales, C=P.
- In total sales, C=F+P

- In BEP sales, C=F
- Below the BEP sales, C=F-P;  C<F
- In M/S sales, C=P
- M/S sales =Total sales-BEP sales
- BEP sales=Total sales-M/S sales
- Total sales=BEP sales+M/S sales




We understood from the above BEP graph that:

- In the above figure, units of products/sales are shown on the horizontal axis OX and costs and revenues are shown on vertical axis OY.
- The variable cost line is drawn first. It increases along with volume of production and sales.
- The total cost line is parallel to variable cost line. It is derived by adding total fixed costs line to the total variable cost line.
- The total revenue line (TR) starts from point (0) and increases along with volume of production or sales intersecting total cost line at point BEP.
- To the right of the BEP is profit zone and to the left of the BEP is the loss zone.
- A perpendicular from the BEP to the horizontal axis at point ‘M’ shows ‘OM’ is the quantity produced at ‘ OP’ the cost at BEP.
- The angle formed by the point of intersection of total revenue and total cost line at BEP is called angle of incidence. The greater the angle of incidence, the higher is the magnitude of profit once the fixed costs are observed.
- Margin of safety refers to the excess of production or sales over and above the BEP. The margin of safety ‘MN’ is the difference between ON and OM (ON-OM=MN). The sales value at ON is OQ.

## SIGNIFICANCE OF BREAK-EVEN ANALYSIS

1) To ascertain the profit on a particular level of sales volume or a given capacity of production.
2) To calculate sales required to earn a particular desired level of profit.
3) To compare the product lines, sales area, method of sale for individual company.
4) To compare the efficiency of the different firms.
5) To decide whether to add a particular product to the existing product line or drop one from it.
6) To decide to ‘make or buy’ a given component or spare part.
7) To decide what promotion mix will yield optimum sales.

8) To assess the impact of changes in fixed cost, variable cost or selling price on BEP and profits during a given period.

## LIMITATIONS OF BREAK-EVEN ANALYSIS

1) Break-even point is based on fixed cost, variable cost and total revenue. A change in one variable is going to affect the BEP.
2) All costs cannot be classified into fixed and variable costs. we have semi-variable costs also.
3) In case of multi-product firm, a single chart cannot be of any use. Series of charts have to be made use of .
4) It is based on fixed cost concept and hence-holds good only in the short-run.
5) Total cost and total revenue lines are not always straight as shown in the figure. The quantity and price discounts are the usual phenomena affecting the total revenue line.
6) Where the business conditions are volatile, BEP cannot give stable results.

## PROBLEMS AND SOLUTIONS:

**1**. A firm has a fixed cost of Rs. 10,000, selling price per unit is Rs.5 and variable cost per unit is Rs. 3.

a. Determine break-even point in terms of volume and also sales value.

b. Calculate the margin of safety considering that the actual production is 8000 units.

**Solution:**

$\boldsymbol{a}. BEP\ (in\ units) = \frac{F}{C\ per\ unit}$

Note: BEP in units can be calculated only when unit sales price and unit variable cost are given.

$$Contribution\ per\ unit = Selling\ price\ per\ unit - Variable\ cost\ per\ unit = S - V = 5 - 3 = 2$$

$$BEP\ (in\ units) = \frac{F}{C\ per\ unit} = \frac{10000}{2} = 5000\ units$$

$$BEP\ (in\ sales) = \frac{F}{P/V\ ratio}$$

$$P/V\ ratio = \frac{C}{S} X\ 100 = \frac{2}{5}\ X\ 100 = 40\%$$

$$BEP\ (in\ sales) = \frac{F}{P/V\ ratio} = \frac{10000}{40\%} = Rs. 25000$$

**(or)**

$$BEP\ (in\ sales) = BEP\ units\ x\ Selling\ Price\ per\ unit = 5000\ x\ 5 = 25000$$

$$b. Margin\ of\ safety\ \boldsymbol{sales} = \frac{P}{P/V\ ratio}$$

Profit = Total Sales – (Total Variable cost + Fixed cost)

Profit = Contribution – Fixed cost

Contribution = Sales – Variable cost

Total Sales = Total Units sold x Selling price per unit = 8000 x 5 = Rs. 40000

Total variable cost x Variable cost per unit = 8000 x 3 = Rs. 24000

Profit = Total Sales – (Total Variable cost + Fixed cost) = 40000 – (24000+10000) = Rs. 6000

$$Margin\ of\ safety\ sales = \frac{P}{P/V\ ratio} = \frac{6000}{40\%} = Rs. 15000$$

(or)

Margin of safety sales = Margin of safety units x Selling price per unit

Margin of safety unit = Total units – BEP units = 8000 – 5000 = 3000

Margin of safety sales = Margin of safety units x Selling price per unit = 3000 x 5 = Rs.15000

**2.** A high-tech rail can carry a maximum of 36,000 passengers per annum at a fare of Rs. 400. The variable cost per passenger is Rs. 150 while the fixed costs are Rs.25,00,000 per year. Find the break-even point in terms of number of passengers and also in terms of fare collections.

**Solution:**

$$\boldsymbol{a}. BEP\ (in\ number\ of\ passengers) = \frac{F}{C\ per\ unit}$$

$$Contribution\ per\ passenger = Selling\ price\ per\ passenger - Variable\ cost\ per\ passenger$$

$$= S - V = 400 - 150 = 250$$

$$BEP\ (in\ number\ of\ passengers) = \frac{F}{C\ per\ unit} = \frac{2500000}{250} = 10000\ passengers$$

$$\boldsymbol{b}. BEP\ (in\ Fare\ collection) = \frac{F}{P/V\ ratio}$$

$$P/V\ ratio = \frac{C}{S} X\ 100 = \frac{250}{400} X\ 100 = 62.50\%$$

$$BEP\ (in\ Fare\ collection) = \frac{F}{P/Vratio} = \frac{2500000}{62.50\%} = \text{Rs. } 4000000$$

**3.** Srikanth Enterprises deals in the supply of hardware parts of computer. The following cost data is available for two successive periods.

| | **Year I (Rs)** | **Year II (Rs)** |
|---|---|---|
| Sales | 50,000 | 1,20,000 |
| Fixed costs | 10,000 | 20,000 |
| Variable cost | 30,000 | 60,000 |

Determine 1. Break-even point, 2. Margin of safety

**Solution:**

1. $BEP\ (in\ sales) = \frac{F}{P/V\ ratio}$

$$\text{P}/V\ ratio = \frac{C}{S} X\ 100 = \frac{S-V}{S} X\ 100$$

$$Year\ I.\ P/V\ ratio = \frac{C}{S} X\ 100 = \frac{S-V}{S} X\ 100 = \frac{50000 - 30000}{50000} X\ 100 = 40\%$$

$$Year\ II.\ P/V\ ratio = \frac{C}{S} X\ 100 = \frac{S-V}{S} X\ 100 = \frac{120000 - 60000}{120000} X\ 100 = 50\%$$

$$Year\ I.\ BEP\ (in\ sales) = \frac{F}{P/V\ ratio} = \frac{10000}{40\%} = Rs. 25000$$

$$Year\ II.\ BEP\ (in\ sales) = \frac{F}{P/V\ ratio} = \frac{20000}{50\%} = Rs. 40000$$

**2.** $Margin\ of\ safety\ \boldsymbol{sales} = \frac{P}{P/Vratio}$

Profit = Total Sales – (Total Variable cost + Fixed cost) = S – (V + F)

Year I. Profit = S – (V + F) = 50000 – (30000+10000) = Rs. 10000

Year II. Profit = S – (V + F) = 120000 – (60000+20000) = Rs. 40000

$$Year\ I.\ Margin\ of\ safety\ \boldsymbol{sales} = \frac{P}{P/Vratio} = \frac{10000}{40\%} = Rs. 25000$$

$$Year\ II.\ Margin\ of\ safety\ \boldsymbol{sales} = \frac{P}{P/Vratio} = \frac{40000}{50\%} = Rs. 80000$$

**4.** From the following data, calculate,

**a**. P/V Ratio, **b**. Profit when sales are Rs. 20,000

Fixed expenses Rs. 4,000, Break-even point Rs. 10,000

**Solution:**

$$\boldsymbol{a}.\ P/V\ ratio = \frac{C}{S} = \frac{S-V}{S} = \frac{F+P}{Total\ Sales} = \frac{F}{BEP\ sales} = \frac{P}{M/S\ sales} = 1 - \frac{V}{S}$$

$$= \frac{Difference\ in\ profits}{Difference\ in\ sales}$$

$$P/V\ ratio = \frac{F}{BEP\ sales} = \frac{4000}{10000} x100 = 40\%$$

**b**. What is profit when total sales are Rs. 20000?.

Total sales = BEP sales + Margin of safety sales

**Note**:- Contribution of total sales includes profit and fixed cost. = C = (P+F)

Contribution of total sales = Total sales x P/V ratio

= 20000 x 40% = 8000

C = (P+F)

P = C – F = 8000-4000 = Rs. 4000

**5.** A company shows the trading results for two periods:

| Period | Sales | Profit |
|---|---|---|
| Period I | Rs. 20,000 | Rs. 1000 |
| Period II | Rs. 10,000 | Rs. 400 |

Calculate P/V Ratio.

**Solution:**

$$P/V\ ratio = \frac{C}{S} = \frac{S-V}{S} = \frac{F+P}{Total\ Sales} = \frac{F}{BEP\ sales} = \frac{P}{M/S\ sales} = 1 - \frac{V}{S}$$

$$= \frac{Difference\ in\ profits}{Difference\ in\ sales}$$

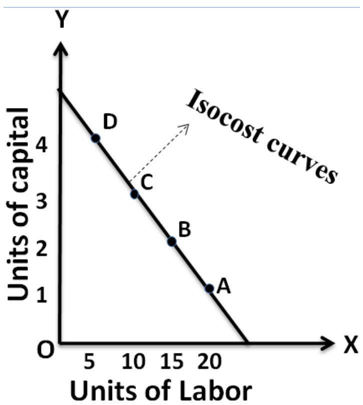
$$P/V\ ratio = \frac{Difference\ in\ profits}{Difference\ in\ sales} = \frac{1000-400}{20000-10000} = \frac{600}{10000} x100 = 6\%$$

## ADDITIONAL INFORMATION

### ISOCOST

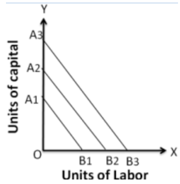
Iso-cost means the line which will show the various combinations of two inputs which can be purchased with a given amount of total money. Suppose total amount is Rs 1000, labor cost is Rs 40 per unit, capital cost is Rs 200 per unit. Alternative combinations are as follows:

| Combination | Capital(K) | Labor(L) | Cost |
|---|---|---|---|
| A | 1 | 20 | 1000 |
| B | 2 | 15 | 1000 |
| C | 3 | 10 | 1000 |
| D | 4 | 5 | 1000 |

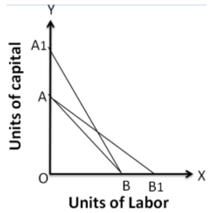


Plotting these values on a graph, joining the points A, B, C, D, then, we will have an isocost line. If the level of production changes, the total cost changes and thus the isocost curve moves upwards, and vice versa.

When outlay is increased, price of factors unchanged, factor combination will change. For each increase in outlay, the isocost lines will be different and shifted to upward and they are parallel till prices of factors are unchanged.




The slope of isocost line vary when factor prices vary. AB is the original isocost line. Total outlay remaining the same, when capital price is lowered, labor price remaining unchanged, the new isocost line A1B is drawn. Similarly, AB1 represents new isocost line when labor price is reduced, capital price being unchanged. AB, A1B and AB1 alternative cost lines corresponding to varying factor prices.




## MARGINAL RATE OF TECHNICAL SUBSTITUTION (MRTS)

The marginal rate of technical substitution (MRTS) refers to the rate at which one input factor can be substituted with the other to attain a given level of output. The slope of isoquant has a technical name called MRTS.

| Combination | Capital(K) | Labor(L) | Production (units) | MRTS |
|---|---|---|---|---|
| A | 1 | 20 | 20000 | - |
| B | 2 | 15 | 20000 | 5:1 |
| C | 3 | 11 | 20000 | 4:1 |
| D | 4 | 8 | 20000 | 3:1 |
| E | 5 | 6 | 20000 | 2:1 |
| F | 6 | 5 | 20000 | 1:1 |

$$\text{MRTS} = \frac{\text{Change in one input}}{\text{Change in another input}} = \frac{\Delta K}{\Delta L}$$

Where ΔK is change in capital and ΔL is change in labor.

$$\text{MRTS} = \frac{2-1}{20-15} = 5:1$$

## LEAST COST COMBINATION OF INPUTS

The manufacturer has to produce at lower costs to attain higher profits. The isocosts and isoquants can be used to determine the inputs usage that minimizes the cost of production.

For example: Here are two input combinations for production of 100 pens. The price of capital and labor are Rs 7 and 10 respectively.

| Combination | Capital(K) | Labor(L) |
|---|---|---|
| A | 10 | 4 |
| B | 8 | 5 |
| C | 6 | 6 |
| D | 5 | 7 |
| E | 4 | 10 |

The following table clarifies the combination which has least cost. The combination no. 3 with a total cost of 102 represents the least cost.

| Combination | Inputs (Units) | | Cost (Rs) | | Total cost (Rs) |
|---|---|---|---|---|---|
| | Capital(K) | Labor(L) | K | L | |
| A | 10 | 4 | 70 | 40 | 110 |
| B | 8 | 5 | 56 | 50 | 106 |
| C | 6 | 6 | 42 | 60 | 102 |
| D | 5 | 7 | 35 | 70 | 105 |
| E | 4 | 10 | 28 | 100 | 128 |

In our two input, one output production function, the cost equation for inputs can be written as C = LPL + KPK where,

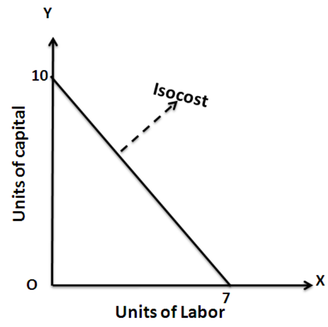
C = Total cost of inputs

L = Quantity of labor

PL = Price of labor

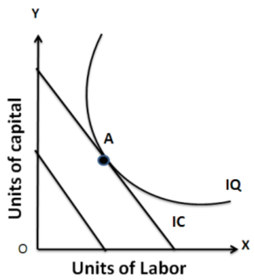
K = Quantity of capital

PK = Price of capital




Thus a linear equation can be obtained for a given level of expenditure and prevailing prices of inputs. For labor costing Rs 10 per unit and capital costing Rs 7 per unit, the cost equation becomes C=10L+7K. If the company has budget of Rs 70, 10L+7K=70 i.e., it can buy7 units of labor with no capital or 10 units of capital with no labor or some in between combination of labor and capital. By joining the two extreme points, we get an isocost.

Once the isocost has been developed for a particular budget, we can draw the isoquant for the required output level on the same graph.




When the isoquant curve is superimposed on the isocost line, the point of tangency between the isoquant and the isocost is the point of the least cost combination of inputs. At that point A, labor and capital are combined in a proportion that maximizes the output for a given budget

.

When a family of isoquants is superimposed on the various possible budget lines(isocost lines), all the isocosts will be tangential. We will thus have a number of least cost combinations one each for every output level. If all the points of tangency between the isocosts and isoquants, representing different output levels are joined, the resultant curve is known as an "Expansion Path". Thus, expansion path is the locus of all possible least cost combinations of inputs for a production function.

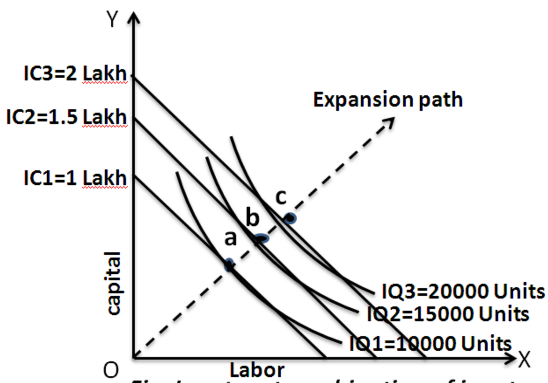


*Fig: Least cost combination of inputs*

The points of tangencies a,b,c, on the isoquant curves represent the least cost combination of inputs yielding maximum level of output.

# ECONOMIES OF SCALE

Production may be carried on a small scale or o a large scale by a firm. When a firm expands its size of production by increasing all the factors, it secures certain advantages known as economies of production. Marshall has classified these economies of large-scale production into internal economies and external economies.

**Internal economies** result from an increase in the scale of output of a firm and cannot be achieved unless output increases. Hence internal economies depend solely upon the size of the firm and are different for different firms.

**External economies** are those benefits, which are shared in by a number of firms or industries when the scale of production in an industry or groups of industries increases. Hence external economies benefit all firms within the industry as the size of the industry expands.

## I. Internal Economies:

### A). Technical Economies.

Technical economies arise to a firm from the use of better machines and superior techniques of production. As a result, production increases and per unit cost of production falls. A large firm, which employs costly and superior plant and equipment, enjoys a technical superiority over a small firm. This increases the productive capacity of the firm and reduces the unit cost of production.

### B). Managerial Economies:

These economies arise due to better and more elaborate management, which only the large size firms can afford. There may be a separate head for manufacturing, assembling, packing, marketing, general administration etc. Each department is under the charge of an expert. Hence the appointment of experts, division of administration into several departments, functional specialization and scientific co-ordination of various works make the management of the firm most efficient.

### C). Marketing Economies:

The large firm reaps marketing or commercial economies in buying its requirements and in selling its final products. The large firm generally has a separate marketing department. It can buy and sell on behalf of the firm, when the market trends are more favorable. In the matter of buying, they could enjoy advantages like preferential treatment, transport concessions, cheap credit, prompt delivery and fine relation with dealers. Similarly it sells its products more effectively for a higher margin of profit.

### D). Financial Economies:

The large firm is able to secure the necessary finances either for fixed capital purposes or for working capital needs more easily and cheaply. It can barrow from the public, banks and other financial institutions at relatively cheaper rates.

### E). Risk bearing Economies:

As there is growth in the size of the firm, there is increase in the risk also. The firm can insure its assets against the hazards of fire, theft and other risks. More often, they deal in more than one product to offset the losses by the profits from the sale of others.

**F). Economies of Research:**

A large firm possesses larger resources and can establish its own research laboratory and employ trained research workers. The firm may even invent new production techniques for increasing its output and reducing cost.

**G). Economies of welfare:**

A large firm can provide better working conditions in-and out-side the factory. Facilities like subsidized canteens, crèches for the infants, recreation room, cheap houses, educational and medical facilities tend to increase the productive efficiency of the workers, which helps in raising production and reducing costs.

**II. External Economies.**

**A). Economies of Concentration:**

When an industry is concentrated in a particular area, all the member firms reap some common economies like skilled labour, improved means of transport and communications, banking and financial services, supply of power and benefits from subsidiaries. All these facilities tend to lower the unit cost of production of all the firms in the industry.

**B). Economies of Information**

The industry can set up an information centre which may publish a journal and pass on information regarding the availability of raw materials, modern machines, export potentialities and provide other information needed by the firms. It will benefit all firms and reduction in their costs.

**C). Economies of Welfare:**

An industry is in a better position to provide welfare facilities to the workers. It may get land at concessional rates and procure special facilities from the local bodies for setting up housing colonies for the workers. It may also establish public health care units, educational institutions both general and technical so that a continuous supply of skilled labour is available to the industry. This will help the efficiency of the workers.

**D). Economies of Disintegration:**

The firms in an industry may also reap the economies of specialization. When an industry expands, it becomes possible to split up some of the processes which are taken over by specialist firms. For example, in the cotton textile industry, some firms may specialize in manufacturing thread, others in printing, still others in dyeing, some in long cloth, some in dhotis, some in shirting etc. As a result, the efficiency of the firms specializing in different fields increases and the unit cost of production falls.

Thus internal economies depend upon the size of the firm and external economies depend upon the size of industry.

## DISECONOMIES OF SCALE

Internal and external diseconomies are the limits to large-scale production. It is possible that expansion of a firm's output may lead to rise in costs and thus result diseconomies instead of economies. When a firm expands beyond proper limits, it is beyond the capacity of the manager to manage it efficiently. This is an example of an internal diseconomy. In the same manner, the expansion of an industry may result in diseconomies, which may be called external diseconomies.

**A). Financial Diseconomies:**

For expanding business, the entrepreneur needs finance. But finance may not be easily available in the required amount at the appropriate time. Lack of finance retards the production plans thereby increasing costs of the firm.

**B). Managerial diseconomies:**

There are difficulties of large-scale management. Supervision becomes a difficult job. Workers do not work efficiently, wastages arise, decision-making becomes difficult, coordination between workers and management disappears and production costs increase.

**C). Marketing Diseconomies:**

As business is expanded, prices of the factors of production will rise. The cost will therefore rise. Raw materials may not be available in sufficient quantities due to their scarcities. Additional output may depress the price in the market. The demand for the products may fall as a result of changes in tastes and preferences of the people. Hence cost will exceed the revenue.

**D). Technical Diseconomies:**

There is a limit to the division of labour and splitting down of production processes. The firm may fail to operate its plant to its maximum capacity. As a result cost per unit increases. Internal diseconomies follow.

**E). Diseconomies of Risk-taking:**

As the scale of production of a firm expands, risks also increase with it. Wrong decision by the management may adversely affect production. If large firms are affected by any disaster, natural or human, the economy will be put to strains.